

EUIJIN HONG

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EDUCATION

Carnegie Mellon University

MS, Electrical and Computer Engineering

Pittsburgh, PA

May 2026

- Cumulative GPA: 4.00/4.00
- Relevant Coursework: Deep Learning Systems (10-714), Optimization (18-660), Advanced Deep Learning (10-707), Deep Generative Modeling (18-789), Introduction to Deep Learning (11-785)

Yonsei University

BS, Electrical and Electronic Engineering

Seoul, South Korea

Feb. 2024

- Major GPA: 3.99/4.00 | Cumulative GPA: 3.91/4.00
- Relevant Coursework: Intelligent Control, Deep Learning Lab, Computer Vision, Introduction to Artificial Intelligence, Probability and Random Variables

PUBLICATIONS

DeltaMomentum: A Key-Value based Anisotropic Momentum Update via Delta Rule

Euijin Hong, Guannan Qu

Under review, NeurIPS 2026 Main Track. Short version at ICML 2026 High-dimensional Learning Dynamics Workshop.

- Replaced the exponential moving average momentum buffer with a delta rule write that reads each gradient as a key and value pair, giving a drop-in substitute for the momentum state of any optimizer at no persistent memory cost.
- Derived the fixed point, the anisotropic memory transport, the per-direction rate reduction, and the non-stationary tracking bound, along with the μP rule under which the single added setting transfers across model widths without re-tuning.
- Reached the AdamW validation loss in up to $46.39 \pm 4.32\%$, $22.12 \pm 0.80\%$, and 19.37% fewer steps at 67M, 370M, and 1B parameters on FineWeb-Edu, with hyperparameter search budget matched across optimizers and a realized per-step overhead of 11.2% and 17.4% at 67M and 370M.
- Traced the gain to the predicted mechanism through training-time diagnostics on momentum tracking and feature covariance conditioning, and showed the same pattern for SGD on CIFAR-10 with ResNet-18 and ViT-Tiny.

STAPLE: Static Sparse Upcycling of Pretrained Checkpoints via Embedding Modules

Ranajoy Sadhukhan, Euijin Hong, Sheng Cao, Zechun Liu, Beidi Chen

Under review, NeurIPS 2026 Main Track.

- Upcycled a pretrained dense checkpoint by turning token-indexed embedding tables into adapters over the feed-forward up-projection, which moves per-token knowledge out of a shared projection and into a sparse lookup.
- Built the gated adapter variant that keeps the pretrained projection and opens a learnable gate onto the embedding path, removing the loss spike that a hard replacement causes at the start of post-training.
- Offloaded the moment states of the vocabulary-sized tables to host memory, which let the adapter runs fit alongside a full Llama-3.2-1B backbone across eight GPUs.
- Measured what the embeddings contribute at the logit level under zero-shot and few-shot evaluation, and compared context-dependent variants of the lookup.

Convolutional Dynamic Rotary Positional Encoding

Euijin Hong, Mengchun Zhang

Interspeech 2026 Main Track.

- Made the position index a function of local acoustic content rather than a fixed grid, generating a bounded offset from a depthwise separable convolution and adding it before rotation, which keeps the rotation harmonically consistent across frequency bands.
- Lowered word error rate on every LibriSpeech split over the relative position baseline, from 5.07% to 4.95% on test-other, using 46.8% fewer learnable positional parameters, with the test-other gain significant under McNemar's test ($p = 0.043$).
- Cut error by up to 19.6% relative under temporal perturbation on Speech Robust Bench, which is where a content-conditioned index is predicted to help, and reported a kernel size sweep as an ablation.

Spectral Equalization Minimizes Total Training Energy: A Control-Theoretic Account of Muon's Advantage

Euijin Hong

ICML 2026 High-dimensional Learning Dynamics Workshop.

- Recast a first-order optimizer as a discrete-time feedback controller and defined total training energy as the closed-loop $\mathcal{H}_2$ norm, which decomposes exactly into per-mode terms under a Kronecker-factored quadratic loss.
- Showed by a Jensen argument that uniform per-mode contraction rates minimize this energy at a fixed mean rate, which is what the polar decomposition step of Muon computes, turning an empirical account of its advantage into a measurable one.
- Released per-mode energy, the Gini coefficient, and the Jensen gap as diagnostics in a monitoring package that attaches to any PyTorch optimizer, and measured the predicted lower per-layer Gini coefficient for Muon on MNIST MLPs.

TALKS AND PRESENTATIONS

Poster, *DeltaMomentum*, ICML 2026 HiLD Workshop, Seoul, South Korea

Jul. 2026

Poster, *Spectral Equalization Minimizes Total Training Energy*, ICML 2026 HiLD Workshop, Seoul, South Korea

Jul. 2026

RESEARCH EXPERIENCE

Hardware-Aware Implementation of DeltaMomentum

Jun. 2026 - Present

Graduate Research Assistant, InfiniAI Lab, Carnegie Mellon University (PI: Prof. Beidi Chen)

- Reordered the correction operator into Gram form and added shape-adaptive per-layer dispatch, an exact rewrite that cut its cost by $7.5\times$ against the naive contraction and took it from 66.7% to 8.9% of forward and backward FLOPs.
- Routed the correction through bf16 tensor cores with fp32 accumulation, raising tensor-core occupancy from zero to full for a $4.2\times$ GEMM speedup, with loss curves indistinguishable from exact fp32.
- Folded the correction statistics into the DDP gradient all-reduce, cutting collectives from 123 to 2 per step and saving 644 MB of traffic per step, and batched the elementwise tail into a single foreach call.
- Held every change to bitwise agreement with a golden reference, and reported the paths that did not pay, including a fused backward overlap whose apparent win was a race artifact and a Triton SYRK kernel dropped on precision grounds.

FFNs with Full-Dimensional Weight Modulation

Jan. 2025 - Jan. 2026

Research Assistant, InfiniAI Lab, Carnegie Mellon University (PI: Prof. Beidi Chen)

- Read the gated linear unit as a local hypernetwork whose gate modulates the base weight along the hidden dimension only, then extended the modulation to a rank-1 outer product acting on both dimensions, raising the degrees of freedom from h to $h + d - 1$ for $O(d^2)$ added parameters.
- Designed and implemented the full architecture family and its training recipe independently before any lab compute was involved, including orthogonal initialization and Muon-based training to control the added gradient paths.
- Pretrained iso-parameter Llama-2 backbones at 125M and 350M, where the modulation lowers perplexity by 0.14 against the gated baseline at 125M and the gap closes at 350M, then diagnosed the closure with effective rank, gradient norm, and Jacobian conditioning probes.

Real-time Neural Network-based Robot Operating System for Teleoperation

Jan. 2023 - Jul. 2023

Undergraduate Researcher, Computational Intelligence Lab, Yonsei University (PI: Prof. Euntai Kim)

- Built a vehicular robot teleoperation system driven by human pose signals, and compressed the YOLOv7-based pose estimation model by branch-wise pruning to cut end-to-end latency while holding pose accuracy.
- Reduced latency by 15.5% at 17.9% fewer parameters while keeping 95.3% teleoperation accuracy.

TEACHING EXPERIENCE

Teaching Assistant

Pittsburgh, PA

Introduction to Deep Learning (11-785), Carnegie Mellon University (Instructor: Prof. Bhiksha Raj)

Aug. 2025 - Present

- Selected for three consecutive semesters, Fall 2025, Spring 2026, and Fall 2026, for a course of 400+ graduate students, and ran office hours, recitations, the class forum, and hackathons across foundational and current topics in deep learning.
- Designed and revised the course assignment on meta-learning across face recognition & face verification and speech-to-text sequence transduction with an encoder-decoder transformer.

HONORS AND AWARDS

Minister's Award

Dec. 2022

Ministry of SMEs and Startups, South Korea

- Awarded to the first-place team in the national Small-Scale Entrepreneur Hackathon competition.

Jilli Scholarship

Mar. 2018 - Sep. 2023

Yonsei University

- Academic scholarship awarded to high academic performers.

High Honors

Mar. 2018, Mar. 2020

Yonsei University

- Granted to the top 3% of students in the academic discipline, among a cohort of 319 students.

SELECTED PROJECTS

Top-K Mixture-of-Experts Language Model via Custom CUDA/C++ Kernels

Oct. 2025 - Dec. 2025

Deep Learning Systems (10-714) Course Project, Carnegie Mellon University

- Added top-k, gather, and scatter to the NDArray backend of the Needle framework in C++ with OpenMP and in CUDA, using partial selection instead of a full sort, then exposed them as differentiable tensor operators.
- Built the router, the SwiGLU experts, and the top-k mixture-of-experts block on those primitives, and trained a full transformer language model end to end inside the framework.

Extending Fourier Number Embeddings to Language Models

Jan. 2025 - Apr. 2025

Introduction to Deep Learning (11-785) Course Project, Carnegie Mellon University

- Designed an injection path that carries Fourier number embeddings into several transformer layers through external encoder networks and parallel adapters, leaving the language modeling objective unchanged.
- Evaluated 6-digit addition and 4-digit multiplication across Fourier, left-to-right, and right-to-left decoding, and identified the mismatch between Fourier encoding and standard tokenizer decoding as the limit on the gain.
- Ranked first in project score among 52 groups, within a cohort of 283 graduate students.

Effective Anomaly Detection Utilizing GAN Discriminators

Jun. 2023 - Jul. 2023

Deep Learning Lab Course Project, Yonsei University

- Built an anomaly detection pipeline that trains two GAN discriminators against separate objectives, one for the generator and one for the detector.
- Reached 99.6% accuracy on in-distribution classification and 96.9% on out-of-distribution detection on MNIST.

TECHNICAL SKILLS

Research Areas: Optimization and Training Dynamics, Neural Network Architecture, Representation Learning, Mechanistic Interpretability, Control Theory, Efficient Training Systems

Programming Languages: Python, CUDA, C/C++, Triton, MATLAB, Verilog, JavaScript

Libraries & Frameworks: PyTorch (DDP, FSDP, custom autograd, torch.compile), HuggingFace, Accelerate, SpeechBrain, Optuna, Weights & Biases, lm-evaluation-harness, NumPy, Pandas

Infrastructure: SLURM-based multi-node training on PSC Bridges-2 and CMU FLAME, kernel-level work on L40 and B200 GPUs

Languages: English (fluent), Korean (native)